

Lab Task

(Raptor Decision Structure)

Graded Lab Task 1 – Water Bill Calculation

Draw a flowchart to calculate the water bill given the cubic feet of water used for a Water Company, which charges the homeowner one of the following:

1. A flat rate of \$15.00 for usage up to and including 1000 cubic feet.
2. \$0.0175 per cubic foot for usage over 1000 cubic feet and up to and including 2000 cubic feet.
3. \$0.02 per cubic foot for usage over 2000 cubic feet and up to and including 3000 cubic feet.
4. A flat rate of \$70.00 for usage over 3000 cubic feet. Test your flowchart with actual data.

Graded Lab Task 2 – Check Digit Algorithm

A company that issues check-cashing cards uses an algorithm to create card numbers. The algorithm adds the digits of a four-digit number, and then adds a fifth digit of 0 or 1 to make the sum of the digits even. The last digit in the number is called the check digit. Draw a flowchart to develop a solution that accepts a four-digit number into one variable, adds the check digit, and prints the original number and the new number.

Test your flowchart with the following data:

Original number = 4737

New number = 47371

And Original Number= 4631

New Number = 46310